LAB EXPERIMENT RECORD

Set 27: Patient Health Risk Analysis

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
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Register Number	192424049
Date	20/10/2026
Day	Tuesday

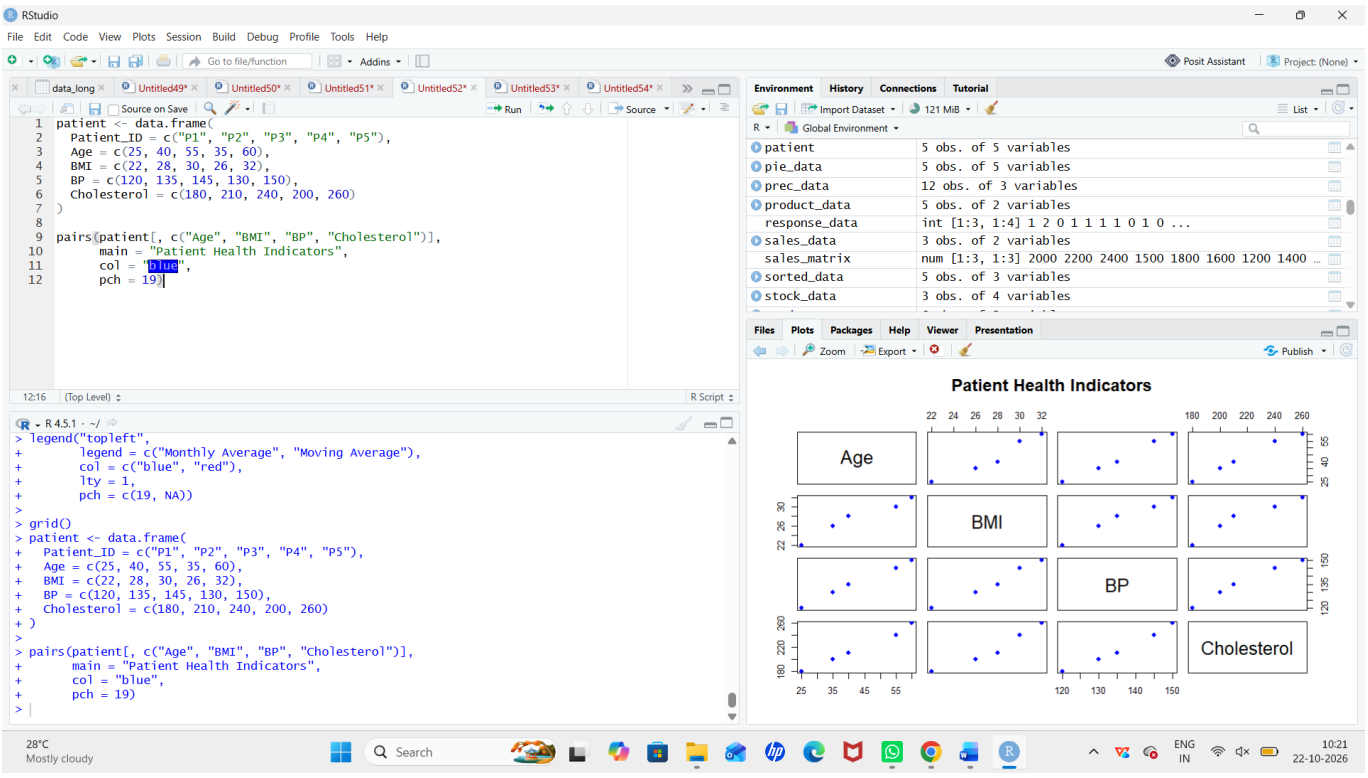
AIM

To analyze patient health data using R and Tableau by creating a scatterplot matrix, Q-Q plot, ECDF, bar chart of average health indicators, and an interactive heatmap dashboard to explore relationships and potential health risk patterns.

1.

```
patient <- data.frame(
  Patient_ID = c("P1", "P2", "P3", "P4", "P5"),
  Age = c(25, 40, 55, 35, 60),
  BMI = c(22, 28, 30, 26, 32),
  BP = c(120, 135, 145, 130, 150),
  Cholesterol = c(180, 210, 240, 200, 260)
)

pairs(patient[, c("Age", "BMI", "BP", "Cholesterol")],
      main = "Patient Health Indicators",
      col = "blue",
      pch = 19)
```



2.

```
cholesterol <- c(180, 210, 240, 200, 260)

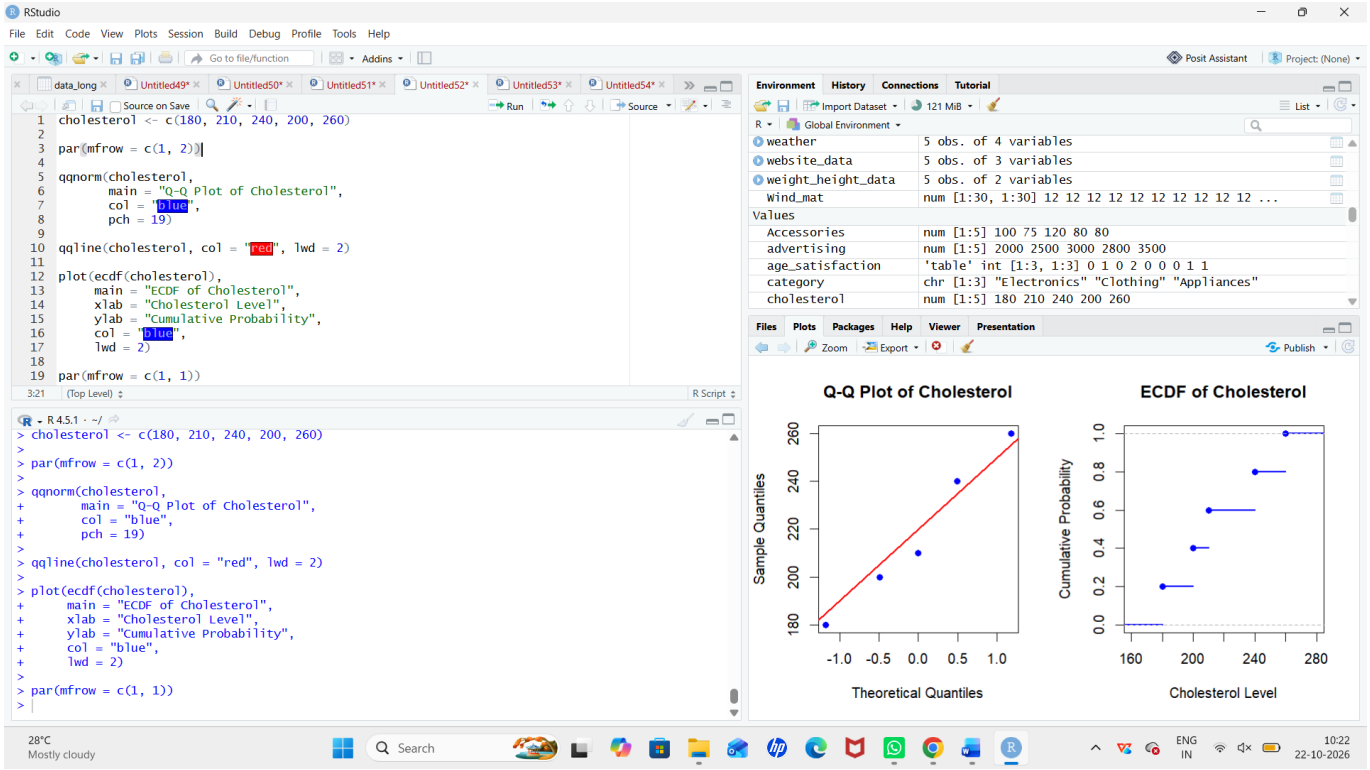
par(mfrow = c(1, 2))

qqnorm(cholesterol,
       main = "Q-Q Plot of Cholesterol",
       col = "blue",
       pch = 19)

qqline(cholesterol, col = "red", lwd = 2)

plot(ecdf(cholesterol),
     main = "ECDF of Cholesterol",
     xlab = "Cholesterol Level",
     ylab = "Cumulative Probability",
     col = "blue",
     lwd = 2)

par(mfrow = c(1, 1))
```



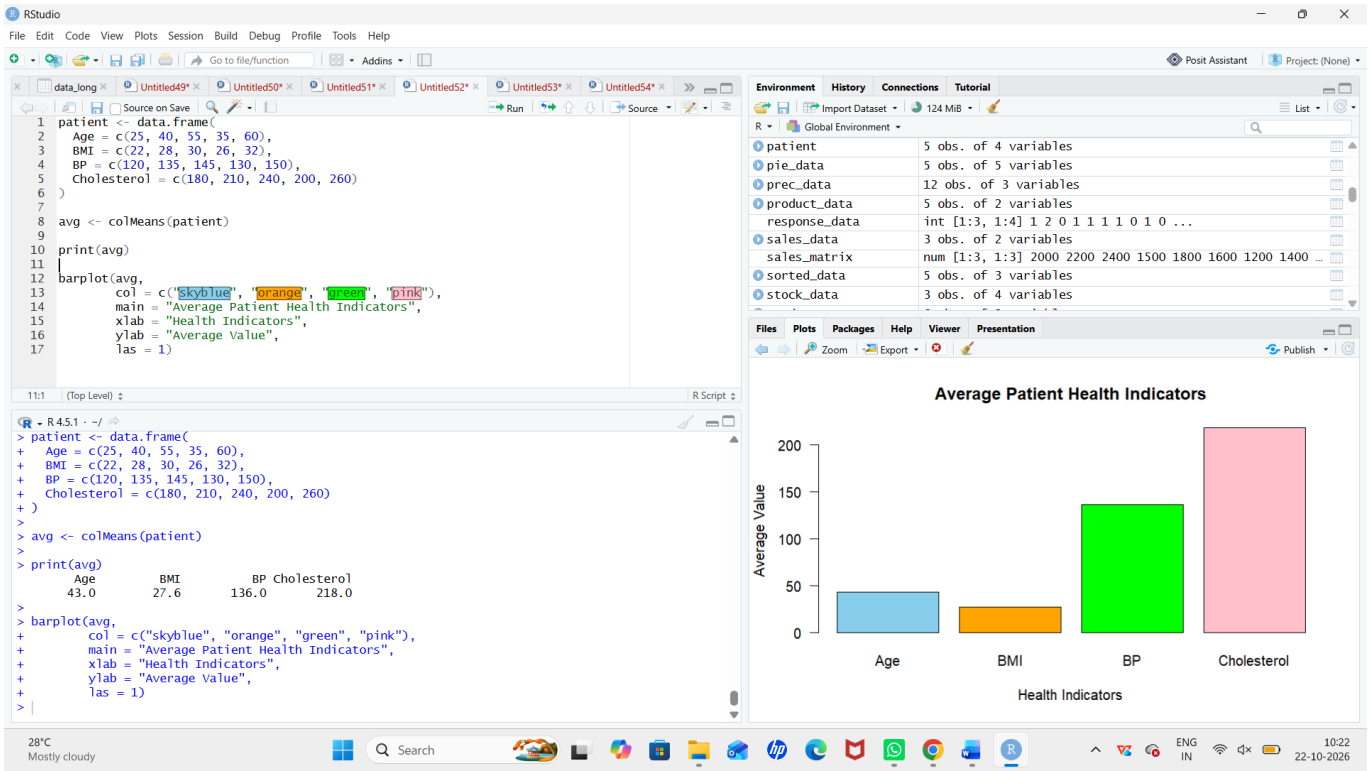
3.

```
patient <- data.frame(
  Age = c(25, 40, 55, 35, 60),
  BMI = c(22, 28, 30, 26, 32),
  BP = c(120, 135, 145, 130, 150),
  Cholesterol = c(180, 210, 240, 200, 260)
)

avg <- colMeans(patient)

print(avg)

barplot(avg,
        col = c("skyblue", "orange", "green", "pink"),
        main = "Average Patient Health Indicators",
        xlab = "Health Indicators",
        ylab = "Average Value",
        las = 1)
```



4.

 Create an Excel or CSV file containing Patient_ID, Age, BMI, BP, and Cholesterol.

 Open Tableau and connect to the dataset.

 Create a worksheet named Patient Health Heatmap.

 Drag Patient_ID to Rows.

 Drag Measure Names to Columns and Measure Values to Color.

 Filter Measure Names to show Age, BMI, BP, and Cholesterol.

 Choose the Square mark type and apply a color gradient to highlight relatively high values.

 Drag Age to Filters and select Show Filter.

 Add Patient_ID, Age, BMI, BP, and Cholesterol to the Tooltip card so these details appear when hovering over a patient.

 Create a dashboard and add the heatmap and Age filter.

 Add a suitable title, such as Patient Health Risk Analysis Dashboard.

Procedure

1. Create the patient dataset containing Patient ID, Age, BMI, Blood Pressure, and Cholesterol.
2. Import and define the dataset in R.
3. Create a scatterplot matrix to examine relationships among the health indicators.
4. Generate a Q-Q plot and ECDF to study the distribution of cholesterol levels.
5. Calculate the average values of the health indicators and display them in a bar chart.
6. Import the dataset into Tableau and create a heatmap using patient IDs and health indicators.

7. Add an Age filter and configure tooltips to display all health measurements.

8. Combine the visualizations into a dashboard and examine patients with relatively high readings.

Result

Patient health data was analyzed successfully using R and Tableau. The scatterplot matrix helped identify relationships among health indicators, while the Q-Q plot and ECDF were used to examine cholesterol distribution. The bar chart displayed average indicator values, and the Tableau dashboard enabled age-based filtering and comparison of individual patients. P5 and P3 showed relatively high values across several indicators, suggesting that further clinical assessment may be appropriate.